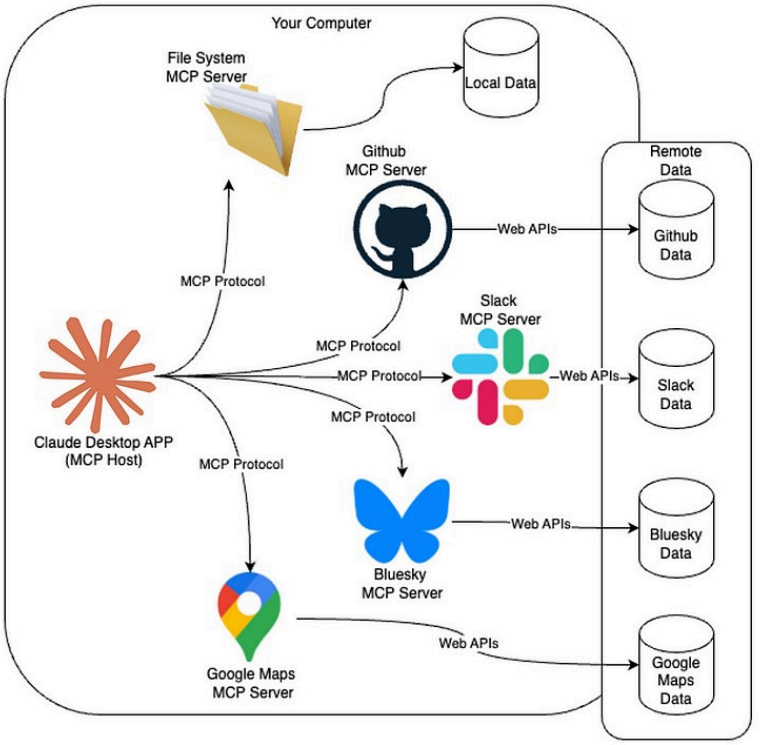
MCP Server

- **MCP = Model Context Protocol**, introduced by **Anthropic**
- It's a standard that **lets AI connect to external tools**
- Turns **APIs into AI-callable tools**
- Instead of AI just chatting, it can fetch real data or **perform actions**
- Example **tools** exposed by **MCP servers:**
  - **Read** files
  - **Query** databases
  - Get **GitHub** issues from GitHub
  - Send messages to **Slack**



How MCP Works ?

- **AI connects** to an **MCP server**
- **Server** lists **available tools**
- **AI chooses** the **relevant tool**
- **MCP server** calls the **API**
- **Result** is **returned** to the **AI**

How to Create a Simple MCP Server ?

1 Install SDK

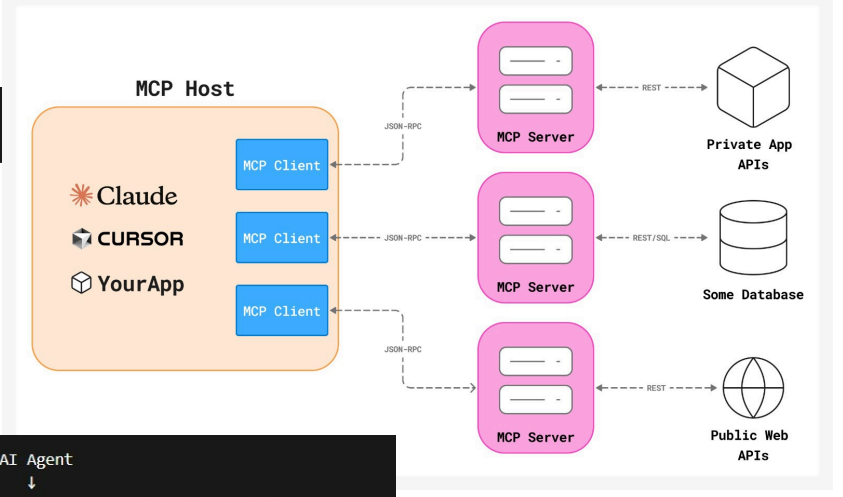
```
npm install @modelcontextprotocol/sdk
```

2 Create a tool

```
server.tool("get_time", async () => {
  return new Date().toString();
});
```

3 Run the server

- AI can now **call get_time() when a user asks for the current time.**




```
AI Agent
  ↓
Connect GitHub MCP → can manage repos
Connect Slack MCP → can send messages
Connect Finance MCP → can analyze revenue
```

Another tool: to get all repos in GitHub account

```
server.tool("get_repositories", async ({ username }) => {
  const res = await fetch(`https://api.github.com/users/${username}/repos`);
  return await res.json();
});
```